SDUHSD CAREER TECHNICAL EDUCATION

Computer Game Design

Campus: Canyon Crest Academy
CTE Industry Sector: Information and Communication Technologies
CTE Pathway: Software and Systems Development
Course Sequence: Intro
Prerequisites: No experience necessary
10 credits, practical art or elective
Recommended Pathway Sequence:

Course 1	Course 2	Course 3	Enrichment
Computer Science Foundations OR Computer Game Design	Human-Computer Interaction (HCI) Design		AP Computer Science Principles AP Computer Science A Machine Learning

Course Description:
Turn your passion for video games, art/graphic design or technology into a future in the multi-billion-dollar video game industry. A wide range of skills will be developed in the class including storyboarding, 3D modeling, rendering, and programming using a variety of software. Once we develop a strong foundation in game design we will focus on creating characters, background maps, and animations to import into various 2D and 3D game editors. This class will be taught through individual/team projects and hands-on assignments. The class culminates with a team game developed over three-weeks and presented to the class. (CCA)

Objectives

- Programming skills: Students will gain a foundational knowledge of programming, basic animation and game design using p5.js and p5Play
- C# Skills: Students will gain a foundational knowledge of programming in C# and will feel confident that they can implement new features on their own with this knowledge
- Unity Skills: Students will have the confidence that, given enough time and resources, they could create anything they want in Unity
- Project Management: As students create their own personal projects, they will learn to manage the process from start to finish: outlining their concept, setting project milestones, and tracking progress.
- Unity Certified User Exam: Should they choose, students will have the skills and confidence to pass the Unity Certified User Programming exam, earning an official certificate validating their skills

- Sketchup for Schools Skills: Model 3D game objects and export them for use in Unity.

Grading:
Course assessment will be mainly project based. Students will learn how to create a given set of structural and design elements in their game in each unit. At the end of each unit, students will submit a project, which will be assessed based on the successful implementation of the required elements from the unit.

Course Syllabi:
CCA - Celniker-GameDesign

Work load: (High, Medium, Low)
Class setting: (This relates directly to how much the student is expected to work independently or in a group requiring self-initiated work, problem solving, etc versus and highly guided and structured class). (Highly independent, Moderately Independent, Structured)

*Note: Each year all CTE courses are reviewed by industry professionals within the pathway to ensure the most current, relevant experience for our students.

Textbooks or Supplemental Materials for this course:

p5.js Library
p5play Library

Software, program knowledge specific to this course:

2D Game Development
3D Game Development with Unity 3D

Units of Study, Large Projects, Work-based Learning Connections.

2D Gaming Principles
2D Game Development
Object-Oriented Programming
3D Game Development with Unity 3D

Industry Connections:

Sony Interactive Entertainment - Development Operations Manager - Sue Xu

The following Model Curriculum Standards connected to this course:

Standards for Career Ready Practice
1. Apply appropriate technical skills and academic knowledge. 4. Apply technology to enhance productivity. 5. Utilize critical thinking to make sense of problems and persevere in solving them. 10. Demonstrate creativity and innovation.

Framework Concept	Software and Systems Development Standards
C1.0 Identify and apply the systems development process.	• C1.1 Identify the phases of the systems development life cycle, including analysis, design, programming, testing, implementation, maintenance, and improvement. • C1.4 Work as a member of, and within the scope and boundaries of, a development project team.
C3.0 Create effective interfaces between humans and technology.	• C3.1 Describe and apply the basic process of input, processing, and output. • C3.2 Design effective and intuitive interfaces using knowledge of cognitive, physical, and social interactions.
C4.0 Develop software using programming languages.	• C4.1 Identify and describe the abstraction level of programming languages from low-level, hardware-based languages to high-level, interpreted, Web-based languages. • C4.3 Identify and use different authoring tools and integrated development environments (IDEs). • C4.7 Use various data structures, arrays, objects, files, and databases. • C4.9 Create programs using control structures, procedures, functions, parameters, variables, error recovery, and recursion. • C4.10 Create and know the comparative advantages of various queue, sorting, and searching algorithms. • C4.11 Document development work for various audiences, such as comments for other programmers, and manuals for users.
C5.0 Test, debug, and improve software development work.	• C5.4 Test software and projects. • C5.5 Evaluate results against initial requirements.
C6.0 Integrate a variety of media into development projects.	• C6.1 Identify the basic design elements necessary to produce effective print, video, audio, and interactive media. • C6.4 Develop a presentation or other multimedia project: video, game, or interactive Web sites, from storyboard to production. • C6.6 Integrate media into a full project using appropriate tools. • C6.7 Create and/or capture professional-quality media, images, documents, audio, and video clips.
C8.0 Develop databases.	• C8.1 Describe the critical function of databases in modern organizations.